



NCI Protocol #: 10323

Patient ID:

Date:

1 of 13

Patient Name:
Date Collected:
Biopsy Site: **Lung**
Sample ID:
Telephone:
Tumor Content (%): **20_49%**

Patient DOB:
Referring Physician:
Primary Diagnosis: **Metastatic colorectal carcinoma**
MoCha ID:
Fax:

Preanalytics to include tumor enrichment as required is performed by Van Andel Research Institute; Grand Rapids; MI 49503

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Relevant Colorectal Cancer Findings

Gene	Finding
BRAF	BRAF V600E
KRAS	None detected
NRAS	None detected
NTRK1	None detected
NTRK2	None detected
NTRK3	None detected

Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IA	BRAF V600E	binimetinib + encorafenib ¹ cetuximab + encorafenib ¹ dabrafenib ¹ dabrafenib + trametinib ¹ trametinib ¹ encorafenib + panitumumab	atezolizumab + cobimetinib + vemurafenib ¹ binimetinib + encorafenib ¹ cetuximab + encorafenib ¹ cobimetinib + vemurafenib ¹ dabrafenib ¹ dabrafenib + trametinib ¹ trametinib ¹ vemurafenib ¹ encorafenib selumetinib	39

Public data sources included in relevant therapies: FDA¹, NCCN

Tier Reference: Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.

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Relevant Biomarkers (continued)

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IIC	PTPRK-RSPO3 fusion	None	None	2
IIC	TP53 R342*	None	None	1

Public data sources included in relevant therapies: FDA¹, NCCN
Tier Reference: Li et al. Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists. J Mol Diagn. 2017 Jan;19(1):4-23.

Variant Details

DNA Sequence Variants							
Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Transcript	Variant Effect
BRAF	p.(V600E)	c.1799T>A	COSM476	chr7:140453136	23.74%	NM_004333.4	missense
TP53	p.(R342*)	c.1024C>T	.	chr17:7574003	28.03%	NM_000546.5	nonsense

Gene Fusions (RNA)		
Genes	Variant ID	Locus
PTPRK-RSPO3	PTPRK-RSPO3.P1R2.COSF1311	chr6:128841403 - chr6:127469792

Comments:
N/A

Report Signed By

Name	Role	Date

Biomarker Descriptions

BRAF (B-Raf proto-oncogene, serine/threonine kinase)

Background: The BRAF gene encodes the B-Raf proto-oncogene serine/threonine kinase, a member of the RAF family of serine/threonine protein kinases which also includes ARAF and RAF1 (CRAF). BRAF is among the most commonly mutated kinases in cancer. Activation of the MAPK pathway occurs through BRAF mutations and leads to an increase in cell division, dedifferentiation, and survival^{1,2}. BRAF mutations are categorized into three distinct functional classes namely, class 1, 2, and 3, and are defined by the dependency on the RAS pathway. Class 1 and 2 BRAF mutants are RAS-independent in that they signal as active monomers (Class 1) or dimers (Class 2) and become uncoupled from RAS GTPase signaling, resulting in constitutive activation of BRAF³. Class 3 mutants are RAS dependent as the kinase domain function is impaired or dead^{3,4,5}.

Alterations and prevalence: Recurrent somatic mutations in BRAF are observed in 40-60% of melanoma and thyroid cancer, approximately 10% of colorectal cancer, and about 2% of non-small cell lung cancer (NSCLC)^{6,7,8,9,10}. Mutations at V600 belong to class

Biomarker Descriptions (continued)

1 and include V600E, the most recurrent somatic BRAF mutation across diverse cancer types^{4,11}. Class 2 mutations include K601E/N/T, L597Q/V, G469A/V/R/, G464V/E/, and BRAF fusions⁴. Class 3 mutations include D287H, V459L, G466V/E/A, S467L, G469E, and N581S/I⁴. BRAF V600E is universally present in hairy cell leukemia, mature B-cell cancer, and prevalent in histiocytic neoplasms^{12,13,14}. Other recurrent BRAF somatic mutations cluster in the glycine-rich phosphate-binding loop at codons 464-469 in exon 11 as well as additional codons flanking V600 in the activation loop¹¹. In primary cancers, BRAF amplification is observed in 8% of ovarian cancer and about 1% of breast cancer^{7,10}. BRAF fusions are mutually exclusive to BRAF V600 mutations and have been described in melanoma, thyroid cancer, pilocytic astrocytoma, NSCLC, and several other cancer types^{15,16,17,18,19}. Part of the oncogenic mechanism of BRAF gene fusions is the removal of the N-terminal auto-inhibitory domain leading to constitutive kinase activation^{5,15,17}.

Potential relevance: Vemurafenib²⁰ (2011) was the first targeted therapy approved for the treatment of patients with unresectable or metastatic melanoma with a BRAF V600E mutation. BRAF class 1 mutations, including V600E, are sensitive to vemurafenib, whereas class 2 and 3 mutations are insensitive⁴. BRAF kinase inhibitors including dabrafenib²¹ (2013) and encorafenib²² (2018) are also approved for the treatment of patients with unresectable or metastatic melanoma with BRAF V600E/K mutations. Encorafenib²² is approved in combination with cetuximab²³ (2020) for the treatment of BRAF V600E mutated colorectal cancer. Due to the tight coupling of RAF and MEK signaling, several MEK inhibitors have been approved for patients harboring BRAF alterations⁴. Trametinib²⁴ (2013) and binimetinib²⁵ (2018) were approved for the treatment of metastatic melanoma with BRAF V600E/K mutations. Combination therapies of BRAF plus MEK inhibitors have been approved in melanoma and NSCLC. The combinations of dabrafenib/trametinib (2015) and vemurafenib/cobimetinib²⁶ (2015) were approved for the treatment of patients with unresectable or metastatic melanoma with a BRAF V600E/K mutation. Subsequently, the combination of dabrafenib and trametinib was approved for metastatic NSCLC (2017) with a BRAF V600E mutation. The PD-L1 antibody, atezolizumab²⁷, has also been approved in combination with cobimetinib and vemurafenib for BRAF V600 mutation-positive unresectable or metastatic melanoma. In 2018, binimetinib²⁸ was also granted breakthrough designation in combination with cetuximab and encorafenib for BRAF V600E mutant metastatic colorectal cancer. The pan-RAF kinase inhibitor, tovorafenib (DAY-101), was granted breakthrough therapy designation (2020) by the FDA for pediatric patients with advanced low-grade glioma harboring activating RAF alterations²⁹. The ERK inhibitor ulixertinib³⁰ was also granted a fast track designation in 2020 for the treatment of patients with non-colorectal solid tumors harboring BRAF mutations G469A/V, L485W, or L597Q. BRAF fusion is a suggested mechanism of resistance to BRAF targeted therapy in melanoma³¹. Additional mechanisms of resistance to BRAF targeted therapy include BRAF amplification and alternative splice transcripts as well as activation of PI3K signaling and activating mutations in KRAS, NRAS, and MAP2K1/2 (MEK1/2)^{32,33,34,35,36,37,38}. Clinical responses to sorafenib and trametinib in limited case studies of patients with BRAF fusions have been reported¹⁹.

RSPO3 (R-spondin 3)

Background: The RSPO3 gene encodes R-spondin 3 protein, a member of the R-spondin family of secreted protein ligands, which includes RSPO1, RSPO2, and RSPO4^{39,40}. R-spondin proteins contribute to the regulation of the Wnt signaling pathway, the activation of which can lead to the expression of genes that control cell proliferation, migration, and cell polarity formation^{40,41,42}. Specifically, Wnt signaling receptors LGR4, LGR5, and LGR6 have been shown to bind to RSPO2 and RSPO3 to enhance Wnt/beta-catenin signaling^{43,44}. Aberrations in the Wnt signaling pathway, including RSPO3 rearrangement, have been observed to lead to Wnt activation, thereby influencing cancer development and progression^{45,46}.

Alterations and prevalence: Rearrangements of RSPO3 that lead to protein fusions are observed to potentiate Wnt signalling and have been identified to be recurrent in colon cancer⁴⁶. In one study, the RSPO3-PTPRK fusion was identified in 5/68 (8%) of colon tumor samples⁴⁶. RSPO3 fusions are also observed in 1% of sarcoma^{7,10}. Somatic RSPO3 mutations are observed in 3% of uterine corpus endometrial carcinoma, 2% of skin cutaneous melanoma, and colorectal adenocarcinoma^{7,10}.

Potential relevance: Currently, no therapies are approved for RSPO3 aberrations.

TP53 (tumor protein p53)

Background: The TP53 gene encodes the p53 tumor suppressor protein that binds to DNA and activates transcription in response to diverse cellular stresses to induce cell cycle arrest, apoptosis, or DNA repair. In unstressed cells, TP53 is kept inactive by targeted degradation via MDM2, a substrate recognition factor for ubiquitin-dependent proteolysis. Alterations in TP53 is required for oncogenesis as they result in loss of protein function and gain of transforming potential⁴⁷. Germline mutations in TP53 are the underlying cause of Li-Fraumeni syndrome, a complex hereditary cancer predisposition disorder associated with early-onset cancers^{48,49}.

Biomarker Descriptions (continued)

Alterations and prevalence: TP53 is the most frequently mutated gene in the cancer genome with approximately half of all cancers experiencing TP53 mutations. Ovarian, head and neck, esophageal, and lung squamous cancers have particularly high TP53 mutation rates (60-90%)^{7,10,50,51,52,53}. Approximately two-thirds of TP53 mutations are missense mutations and several recurrent missense mutations are common including substitutions at codons R158, R175, Y220, R248, R273, and R282^{7,10}. Invariably, recurrent missense mutations in TP53 inactivate its ability to bind DNA and activate transcription of target genes^{54,55,56,57}.

Potential relevance: The small molecule p53 reactivator, PC14586, received a fast track designation (2020) by the FDA for advanced tumors harboring a TP53 Y220C mutation⁵⁸. The FDA has granted fast track designation (2019) to the p53 reactivator, eprenetapopt,⁵⁹ and breakthrough designation⁶⁰ (2020) in combination with azacitidine or azacitidine and venetoclax for acute myeloid leukemia patients (AML) and myelodysplastic syndrome (MDS) harboring a TP53 mutation, respectively. In addition to investigational therapies aimed at restoring wild-type TP53 activity, compounds that induce synthetic lethality are also under clinical evaluation^{61,62}. TP53 mutations confer poor prognosis in multiple blood cancers including AML, MDS, myeloproliferative neoplasms (MPN), and chronic lymphocytic leukemia (CLL)^{63,64,65,66}. In mantle cell lymphoma, TP53 mutations are associated with poor prognosis when treated with conventional therapy including hematopoietic cell transplant⁶⁷. Mono- and bi-allelic mutations in TP53 confer unique characteristics in MDS, with multi-hit patients also experiencing associations with complex karyotype, few co-occurring mutations, and high-risk disease presentation as well as predicted death and leukemic transformation independent of the IPSS-R staging system⁶⁸.

Relevant Therapy Summary

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

BRAF V600E

Relevant Therapy	FDA	NCCN	Clinical Trials*
cetuximab + encorafenib	☒	☒	☒
binimetinib + encorafenib	☒	☐	☒
dabrafenib	☒	☐	☒
dabrafenib + trametinib	☒	☐	☒
trametinib	☒	☒	☒
atezolizumab + cobimetinib + vemurafenib	☐	☐	☒ (II)
cobimetinib + vemurafenib	☐	☐	☒ (II)
vemurafenib	☐	☐	☒ (II)
encorafenib + panitumumab	☒	☒	☒
encorafenib	☒	☐	☒
selumetinib	☒	☐	☒
bevacizumab, chemotherapy	☒	☒	☒ (III)
encorafenib, cetuximab, chemotherapy	☒	☒	☒ (III)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

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Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

BRAF V600E (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
nivolumab, encorafenib, binimetinib, cetuximab	×	×	● (III)
cetuximab, binimetinib, encorafenib	×	×	● (II/III)
camrelizumab, regorafenib, fruquintinib	×	×	● (II)
cetuximab, vemurafenib, chemotherapy	×	×	● (II)
dabrafenib, trametinib, spartalizumab	×	×	● (II)
encorafenib, binimetinib, cetuximab	×	×	● (II)
encorafenib, cetuximab, nivolumab	×	×	● (II)
panitumumab, trametinib	×	×	● (II)
RX208, cetuximab	×	×	● (II)
vemurafenib, cobimetinib	×	×	● (II)
ASN007, encorafenib, cetuximab	×	×	● (I/II)
OKI-179, binimetinib	×	×	● (I/II)
RX208, trametinib	×	×	● (I/II)
vemurafenib, cetuximab, chemotherapy	×	×	● (I/II)
ABM-1310	×	×	● (I)
binimetinib	×	×	● (I)
CGX-1321, encorafenib, cetuximab	×	×	● (I)
cobimetinib, belvarafenib, cetuximab	×	×	● (I)
dabrafenib, LTT-462, trametinib, LXH254 , TNO-155, spartalizumab	×	×	● (I)
HL-085, vemurafenib	×	×	● (I)
JSI-1187, dabrafenib	×	×	● (I)
LGK-974	×	×	● (I)
PF 07284890, binimetinib	×	×	● (I)
PF-07284892, encorafenib, cetuximab	×	×	● (I)
RO-7276389, cobimetinib	×	×	● (I)

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Relevant Therapy Summary (continued)

☒ In this cancer type

☐ In other cancer type

☒ In this cancer type and other cancer types

☒ No evidence

BRAF V600E (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
RX208	×	×	● (I)
vemurafenib, cetuximab, camrelizumab	×	×	● (I)
ZEN-3694, binimetinib	×	×	● (I)
zimberelimab, pimitespib	×	×	● (I)

PTPRK-RSPO3 fusion

Relevant Therapy	FDA	NCCN	Clinical Trials*
CGX-1321, encorafenib, cetuximab	×	×	● (I)
LGK-974	×	×	● (I)

TP53 R342*

Relevant Therapy	FDA	NCCN	Clinical Trials*
niraparib	×	×	● (II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Clinical Trials Summary

BRAF V600E + PTPRK-RSPO3 fusion

NCT ID	Title	Phase
NCT02675946	A Phase I Open-label Dose Escalation Study of CGX1321 in Subjects With Advanced Solid Tumors With Expansion in Advanced Gastrointestinal Tumors and Phase 1b Study of CGX1321 in Combination With Pembrolizumab in Subjects With Advanced Colorectal Cancer or in Combination With Encorafenib + Cetuximab in Subjects With BRAFV600E Mutated Advanced Colorectal Cancer	I
NCT01351103	A Phase I, Open-label, Dose Escalation Study of Oral LGK974 in Patients With Malignancies Dependent on Wnt Ligands.	I

Clinical Trials Summary (continued)

BRAF V600E

NCT ID	Title	Phase
NCT04607421	An Open-label, Multicenter, Randomized Phase III Study of First Line Encorafenib plus Cetuximab with or Without Chemotherapy Versus Standard of Care Therapy with a Safety Lead-in of Encorafenib and Cetuximab plus Chemotherapy in Participants with Metastatic BRAF V600E Mutant Colorectal Cancer.	III
No NCT ID	OGSG2101: Stage II / III Investigating the Efficacy and Safety of Encorafenib + Binimetinib + Cetuximab Therapy for Patients with RAS Wild-type and BRAF V600E Mutation-recurrent Colorectal Cancer who Relapsed Early During or After Adjuvant Chemotherapy After Radical Resection of Colorectal Cancer. Phase II study (TRESBIEN study)	II/III
No NCT ID	Camrelizumab Combined with Fruquintinib or Regorafenib in The Treatment of Second-Line And Later-Line BRAF Mutation-Positive, Microsatellite Stable Advanced Colorectal Cancer: A Prospective Clinical Study	II
NCT03727763	A Single Center Clinical Study of FOLFIRI With Vemurafenib and Cetuximab in Advanced Colorectal Cancer Patients With BRAF V600E Mutation.	II
NCT03668431	A Phase II Study of Dabrafenib and Trametinib in Combination With PDR001 in Patients With BRAFV600E Metastatic Colorectal Cancer	II
NCT04984369	An Open, Multicenter Phase II Clinical Study to Evaluate Safety and Efficacy of HLX208 (BRAF V600E Inhibitor) Combined With Cetuximab in Patients With Metastatic Colorectal Cancer (mCRC)	II
NCT05127759	An Open Label, Multicenter Phase II Clinical Study to Evaluate Efficacy , Safety and PK of HLX208 in Metastatic Colorectal Cancer (mCRC) With BRAF V600E Mutation	II
NCT04591431	The Rome Trial From Histology to Target: the Road to Personalize Target Therapy and Immunotherapy	II
NCT05039177	A Phase Ib/II Study of Agents Targeting the Mitogen-Activated Protein Kinase Pathway in Patients With Advanced Gastrointestinal Malignancies (HERKULES-3)	I/II
NCT04790448	A Phase Ia/II, Single Arm Trial on the Efficacy of Vemurafenib in Combination With Irinotecan and Cetuximab in BRAF V600E-Mutant Metastatic Colorectal Cancer	I/II
NCT04190628	A Phase I, First-In-Human, Multicenter, Open-Label Study of ABM-1310, Administered Orally in Adult Patients With Advanced Solid Tumors	I
NCT04800822	A Phase I Open-Label, Multi-Center, Dose Escalation and Dose Expansion Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Preliminary Evidence of Antitumor Activity of PF-07284892 (ARRY-558) as a Single Agent and in Combination Therapy in Participants With Advanced Solid Tumors	I
NCT05004350	A Multicenter, Randomized, Open-label, 2-arm, Phase II Study With a Safety lead-in Phase Evaluating the Combination of Encorafenib and Cetuximab Versus Irinotecan/Cetuximab or Infusional 5-fluorouracil (5-FU)/Folinic Acid (FA)/Irinotecan (FOLFIRI)/Cetuximab in Chinese Patients With BRAF V600E Mutant Metastatic Colorectal Cancer.	II
NCT03087071	A Phase II Enrichment Study of Panitumumab as a Single Agent or in Combination With Trametinib in Anti-EGFR-Refractory Stage IV Colorectal Cancer Patients	II
NCT03284502	A Phase Ib, Open-label, Multicenter, Dose Escalation Study of the Safety, Tolerability, and Pharmacokinetics of HM95573 in Combination With Either Cobimetinib or Cetuximab in Patients With Locally Advanced or Metastatic Solid Tumors	I
NCT04999761	Platform Study of AB122 Based Treatments in Patients With Advanced Solid Tumors	I

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Clinical Trials Summary (continued)

BRAF V600E (continued)

NCT ID	Title	Phase
NCT03803553	Early Identification and Treatment of Occult Metastatic Disease in Stage III Colon Cancer	III
NCT04551521	Continuous ReAssessment With Flexible ExTension in Rare Malignancies - CRAFT: The NCT-PMO-1602 Phase II Trial	II
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study	II
No NCT ID	A Phase II Study of Encorafenib + Binimetinib + Cetuximab in Patients with BRAF V600E-mutant Metastatic Colorectal Cancer After Refractory to Encorafenib + Cetuximab (BAYONET)	II
No NCT ID	A Phase II Study of Rechallenge with Encorafenib + Binimetinib + Cetuximab in Patients with BRAF V600E-mutant Metastatic Colorectal Cancer with Prior Treatment History of Combination Therapy Including Encorafenib + Cetuximab (TRIDENTE)	II
NCT05308446	Randomized Phase II Trial of Encorafenib and Cetuximab With or Without Nivolumab (NSC #748726) for Patients With Previously Treated, Microsatellite Stable, BRAFV600E Metastatic and/or Unresectable Colorectal Cancer	II
No NCT ID	A Phase IA/B Open-label Study to Evaluate Safety, Pharmacokinetics, and Preliminary Clinical Activity of R07276389 alone and in Combination with Cobimetinib in Participants with BRAF-V600 Mutation-positive Advanced Solid Tumor or BRAF-V600 Mutation-positive Melanoma with Central Nervous System Metastases	I
No NCT ID	Phase I study of the safety, tolerability, pharmacokinetics and preliminary efficacy of RX208 in patients with advanced malignant solid tumors	I
NCT05019534	A Phase I Study on Tolerance and Safety of Vemurafenib Film-coated Tablets, Cetuximab Solution for Infusion and Camrelizumab Protocol(VCC) in the After Line Therapy of BRAF V600E Mutation/MSS Metastatic Colorectal Cancer	I
No NCT ID	JCOG2014: Randomized Phase III Study of Bi-weekly Trifluridine/Tipiracil Plus Bevacizumab Vs Trifluridine/Tipiracil Alone for Chemorefractory Metastatic Colorectal Cancer (ROBiTS)	III
NCT03297606	Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR): A Phase II Basket Trial	II
No NCT ID	Single Arm, Open Label, Phase II Trial Of Vemurafenib And Cobimetinib In Patients With Metastatic Non-Squamous Non-Small Cell Lung Cancer And Other Tumours Harboursing BRAF V600 Mutations Detected By Comprehensive Genomic Profiling	II
NCT04965220	A Phase Ib/II Clinical Trial Evaluating the Safety, Tolerability, Pharmacokinetics, and Initial Efficacy of HLX208 (BRAF V600E Inhibitor) in Combination With Trametinib in Patients With Advanced Solid Tumors	I/II
NCT04294160	A Phase Ib, Multicenter, Open-label Dose Escalation and Expansion Platform Study of Select Drug Combinations in Adult Patients With Advanced or Metastatic BRAF V600 Colorectal Cancer	I
NCT03781219	A Phase I, Single Arm, Dose Escalation Study to Evaluate Safety, Pharmacokinetics and Preliminary Efficacy of HL-085 Plus Vemurafenib in Patients With BRAF V600 Mutant Advanced Solid Tumor	I
NCT04418167	A Phase I Study of ERK1/2 Inhibitor JSI-1187 Administered as Monotherapy and in Combination With Dabrafenib for the Treatment of Advanced Solid Tumors With MAPK Pathway Mutations	I

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Clinical Trials Summary (continued)

BRAF V600E (continued)

NCT ID	Title	Phase
NCT04543188	A Two-Part, Phase Ia/b, Open-Label, Multicenter Trial Evaluating Pharmacokinetics, Safety and Efficacy of PF 07284890 (ARRY 461) in Participants with Braf V600 Mutant Solid Tumors with and without Brain Involvement	I
NCT05340621	A Phase Ib/II Study of OKI-179 Plus Binimetinib in Patients With Advanced Solid Tumors and Activating Mutations in the RAS Pathway (Phase 1b) and in Patients With Advanced NRAS-Mutated Melanoma (Phase II)	I/II
NCT05111561	A Phase I Study of ZEN003694 in Combination With Binimetinib in Solid Tumors With RAS Pathway Alterations and Triple Negative Breast Cancer	I
NCT04423185	Platform Study of Genotyping Guided Precision Medicine for Rare Tumors in China	II
NCT04913285	A Phase 1/1b Open-label, Multicenter Study to Investigate the Safety, Tolerability, Pharmacokinetics, and Antitumor Activity of KIN-2787 in Participants With BRAF and/or NRAS Mutation-positive Solid Tumors.	I

TP53 R342*

NCT ID	Title	Phase
NCT04992013	Genomically Guided Phase II Study to Evaluate the Clinical Benefit of Niraparib in Tumors Metastatic to the CNS	II

Assay Information

Methodology, Scope, and Application: The OCAv3 next-generation sequencing (NGS) assay identifies more than 3000 annotated mutations of interest (MOIs) broadly categorized into 5 mutation types: single nucleotide variants (SNVs), small insertions/deletions (Indels), large (>3 bases) insertions/deletions (Large Indels), copy number variants (CNVs), and gene fusions. The assay utilizes the Thermo Fisher Scientific Oncomine® Comprehensive Assay v3.0 (OCAv3), a next-generation sequencing (NGS) assay that utilizes a multiplex polymerase chain reaction (PCR) with DNA and RNA extracted from formalin-fixed tissue for sequencing on the Ion S5 XL platform and analyzed by the current version of Torrent Suite Software and Ion Reporter. The OCAv3 NGS Assay currently can reliably identify the presence or absence of >3000 known mutations and polymorphisms in the 161 unique genes compared to the Human Reference Genome hg19, including the genes listed below. The OCAv3 assay is a laboratory developed test designed to find gene mutations within tumors (somatic mutations).

Analytical Sensitivity and Specificity: The OCAv3 NGS Assay has been determined to be suitably analytically sensitive and specific for the various types of abnormalities within its reportable range, demonstrating 97.18% sensitivity compared with orthogonal assay results and 100% specificity for the 5 mutation types reported. 99.99% or greater reproducibility has been demonstrated. All quality measures for this assay were within defined assay parameters.

Hereditary and Germline Mutations: This assay examines tumor tissue only and does not examine normal (non-tumor) tissue. Mutations detected by the assay may be present only in the tumor, or in every cell of the body (including non-tumor cells). This test also cannot tell whether a potential germline mutation causes or will cause a hereditary cancer syndrome. If the patient's history includes any one or combination of features suggestive of a possible hereditary cancer predisposition (such as, cancer arising at a young age, especially a cancer of a type unusual in younger patients; a personal history of multiple different types of cancers; or a significant cancer history among blood relatives, especially but not exclusively of the same type as the patient's) it is recommended that the physician arrange for the patient to meet with a genetic counselor and, if warranted, undergo the appropriate genetic test on normal

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(i.e. non-tumor) tissue (blood or cells brushed from the oral surface of the cheek) to check for a germline abnormality, regardless of the results of this research study.

Report Content: Thermo Fisher Scientific's Ion Torrent OncoPrint Reporter software was used in the generation of this report. Software was developed and designed internally by Thermo Fisher Scientific. The analysis was based on the current version of OncoPrint Reporter. The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. FDA information was sourced from www.fda.gov and is current as of this version. NCCN information was sourced from www.nccn.org and reflects its current version. Clinical Trials information reflects the current version. For the most up-to-date information regarding a particular trial, search www.clinicaltrials.gov by NCT ID or search local clinical trials authority website by local identifier listed in 'Other Identifiers.' Variants are reported according to HGVS nomenclature and classified following AMP/ASCO/CAP guidelines (Li et al. 2017). Based on the data sources selected, variants, therapies, and trials identified in this report are listed in order of potential clinical significance but not for predicted efficacy of the therapies.

Genes assayed for hotspot mutations:

AKT1, AKT2, AKT3, ALK, AR, ARAF, AXL, BRAF, BTK, CBL, CCND1, CDK4, CDK6, CHEK2, CSF1R, CTNNB1, DDR2, EGFR, ERBB2, ERBB3, ERBB4, ERCC2, ESR1, EZH2, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, FOXL2, GATA2, GNA11, GNAQ, GNAS, H3F3A, HIST1H3B, HNF1A, HRAS, IDH1, IDH2, JAK1, JAK2, JAK3, KDR, KIT, KNSTRN, KRAS, MAGOH, MAP2K1, MAP2K2, MAP2K4, MAPK1, MAX, MDM4, MED12, MET, MTOR, MYC, MYCN, MYD88, NFE2L2, NRAS, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPP2R1A, PTPN11, RAC1, RAF1, RET, RHEB, RHOA, ROS1, SF3B1, SMAD4, SMO, SPOP, SRC, STAT3, TERT, TOP1, U2AF1, XPO1

Genes assayed with full exon coverage:

ARID1A, ATM, ATR, ATRX, BAP1, BRCA1, BRCA2, CDK12, CDKN1B, CDKN2A, CDKN2B, CHEK1, CREBBP, FANCA, FANCD2, FANCI, FBXW7, MLH1, MRE11, MSH2, MSH6, NBN, NF1, NF2, NOTCH1, NOTCH2, NOTCH3, PALB2, PIK3R1, PMS2, POLE, PTCH1, PTEN, RAD50, RAD51, RAD51B, RAD51C, RAD51D, RB1, RNF43, SETD2, SLX4, SMARCA4, SMARCB1, STK11, TP53, TSC1, TSC2

Genes assayed for copy number variations:

AKT1, AKT2, AKT3, ALK, AR, AXL, BRAF, CCND1, CCND2, CCND3, CCNE1, CDK2, CDK4, CDK6, EGFR, ERBB2, ESR1, FGF19, FGF3, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, IGF1R, KIT, KRAS, MDM2, MDM4, MET, MYC, MYCL, MYCN, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPARG, RICTOR, TERT

Genes assayed for detection of fusions:

AKT2, ALK, AR, AXL, BRAF, BRCA1, BRCA2, CDKN2A, EGFR, ERBB2, ERBB4, ERG, ESR1, ETV1, ETV4, ETV5, FGFR1, FGFR2, FGFR3, FGF, FLT3, JAK2, KRAS, MDM4, MET, MYB, MYBL1, NF1, NOTCH1, NOTCH4, NRG1, NTRK1, NTRK2, NTRK3, NUTM1, PDGFRA, PDGFRB, PIK3CA, PPARG, PRKACA, PRKACB, PTEN, RAD51B, RAF1, RB1, RELA, RET, ROS1, RSPO2, RSPO3, TERT

DISCLAIMER

This assay is considered a Laboratory Developed Test (LDT). Its performance characteristics have been determined through extensive testing although it has not been cleared by the US Food and Drug Administration and such approval is not required for clinical implementation. Furthermore, any comments included in the report are strictly interpretive and the opinion of the reviewer. This laboratory is certified under the Clinical Laboratory Improvements Amendments (CLIA) for the performance of high-complexity testing for clinical purposes. The correlative data presented is a result of the curation of published data sources, but may not be exhaustive. The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

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